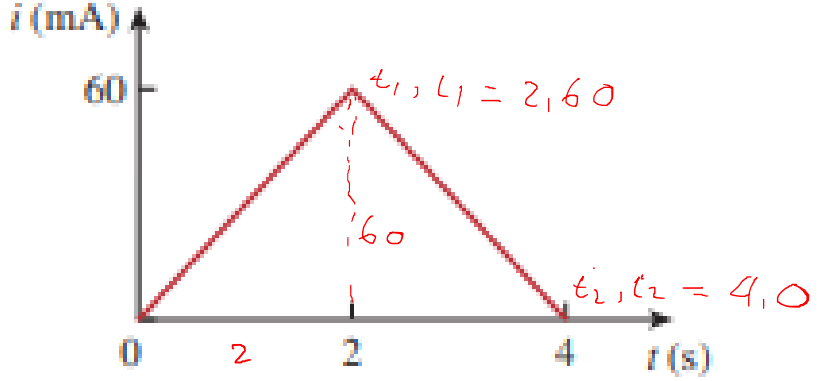
Energi → i ⇒ v i t

⇒ $0 < t < 4$



$0 < t < 2$

$m = \frac{60}{2} = 30$ → persamaan garis (x_1, y_1)

$y - y_1 = m(x - x_1)$ ⇒ $y = i$, $x = t$

$i = 0$ & $t = 0$

$i - 0 = 30(t - 0)$

$i_1(t) = 30t$ mA ... ①

$m =$

Was $\Delta \Rightarrow \frac{4 \times 60}{2} = 120$

$2 < t < 4$

$m = \frac{i_1 - i_2}{t_1 - t_2} = \frac{i_2 - i_1}{t_2 - t_1} = \frac{60 - 0}{2 - 4} = -30$

titik = 4,0

$i - 0 = -30(t - 4)$

$i_2 = -30t + 120$ mA

$0 < t < 4$

$i(t) = i_1(t) + i_2(t)$

$= 30t - 30t + 120$

$= 120$ mA.

(a)

$E = V i t$

$= 120$ V. t mjoule

6

titik A :

$i_{R4} = 4 + 2 = 6$ A

$V_{R4} = 4 \cdot 6 = 12$ V

$V_{AB} = 12$ V

titik BC

$V_{BC} = 2 \times 3 = 6$ V

titik DC

$i_{DC} + 3 = 7$

$i_{DC} = 4$ A.

$V_{DC} = 4 \cdot 2 = 8$ Volt

$V_1 = V_{DC} = 8$ Volt

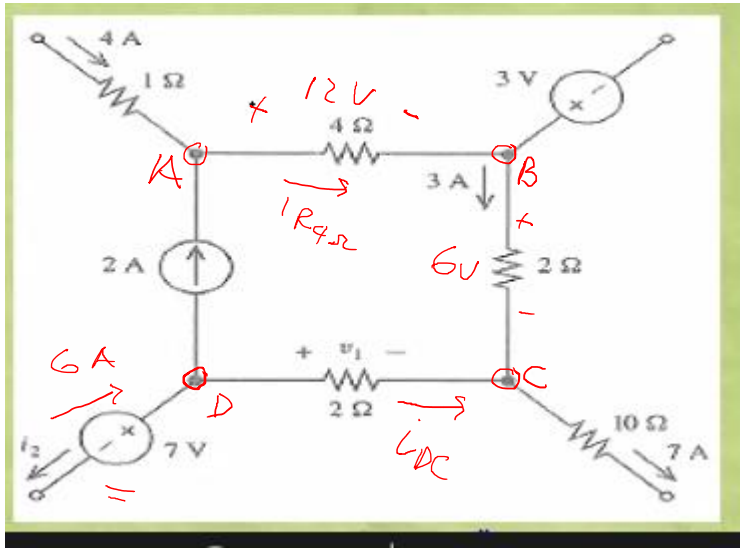
titik D

$2 + 4 + i_2 = 0$

$i_2 = -6$ A.

Sumber tegangan 7 V

- Deliver ✓
- Absorb



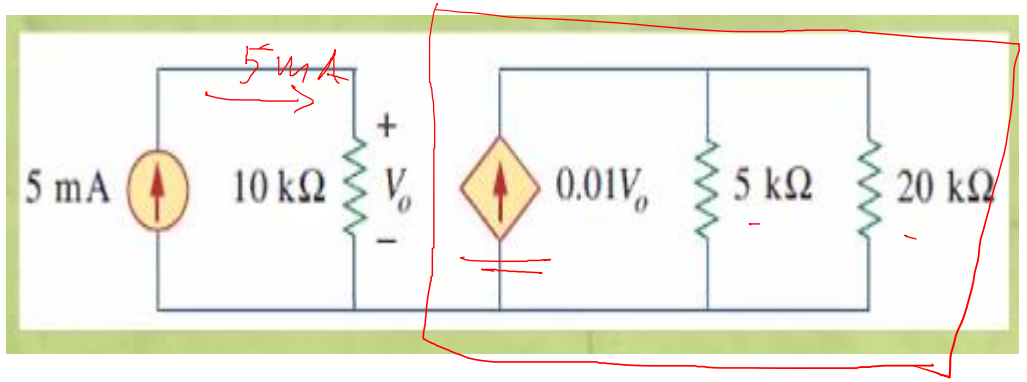
2. ①

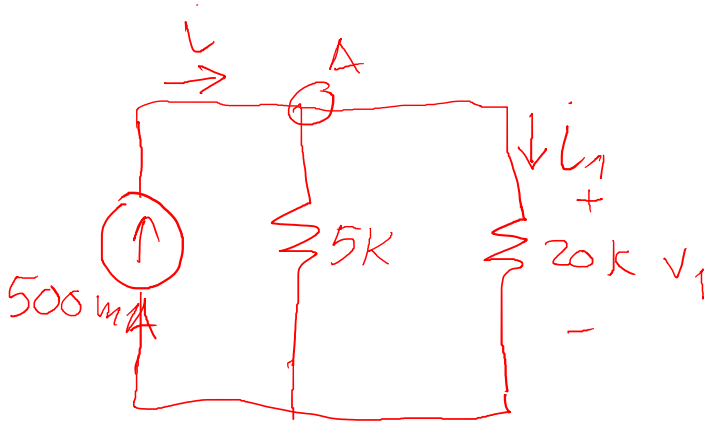
$V_0 = 10\,10^{3} \times 5.10^{-3}$

$= 50\,V$

Sumber Arus $\Rightarrow I = 0{,}01 \times 50$

$= 0{,}5\,A = 500\,mA.$





Pembagi Arus titik A.

$i_1 = \frac{5}{5+20} \times 500 = 100\ mA$

$V_1 = 100\,10^{-3} \times 20\,10^{3}$

$= 2000\ Volt$

$= 2\ k\,Volt$

2. (6)

$i_3 = -16\,A$

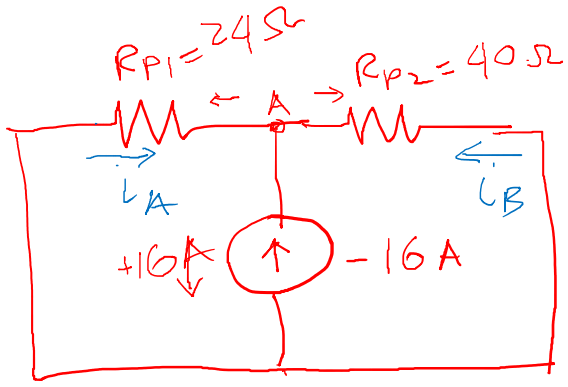
$i_{R_{P1}} = \frac{R_{P2}}{R_{P1}+R_{P2}} \times (-16\,A)$

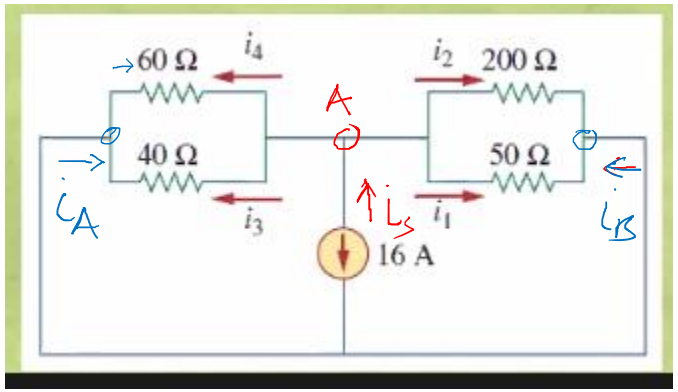
$= \frac{40}{64} \times (-16) = -10\,A$

$i_4 = \frac{40}{40+60} \times (-10) = -4\,A$

$i_{R_{P2}} = \frac{24}{64}(-16) = -6\,A$

$i_1 = \frac{200}{200+50} \times (-6) = -4{,}8\,A$





$R_{P1} = \frac{60 \times 40}{60+40} = \frac{2400}{100} = 24\,\Omega$

$R_{P2} = \frac{200 \times 50}{200+50} = \frac{10000}{250} = 40\,\Omega$

$i_A = \frac{R_{P2}}{R_{P1}+R_{P2}} * (16\,A)$

$= \frac{40}{64} \times 16\,A = 10\,A$

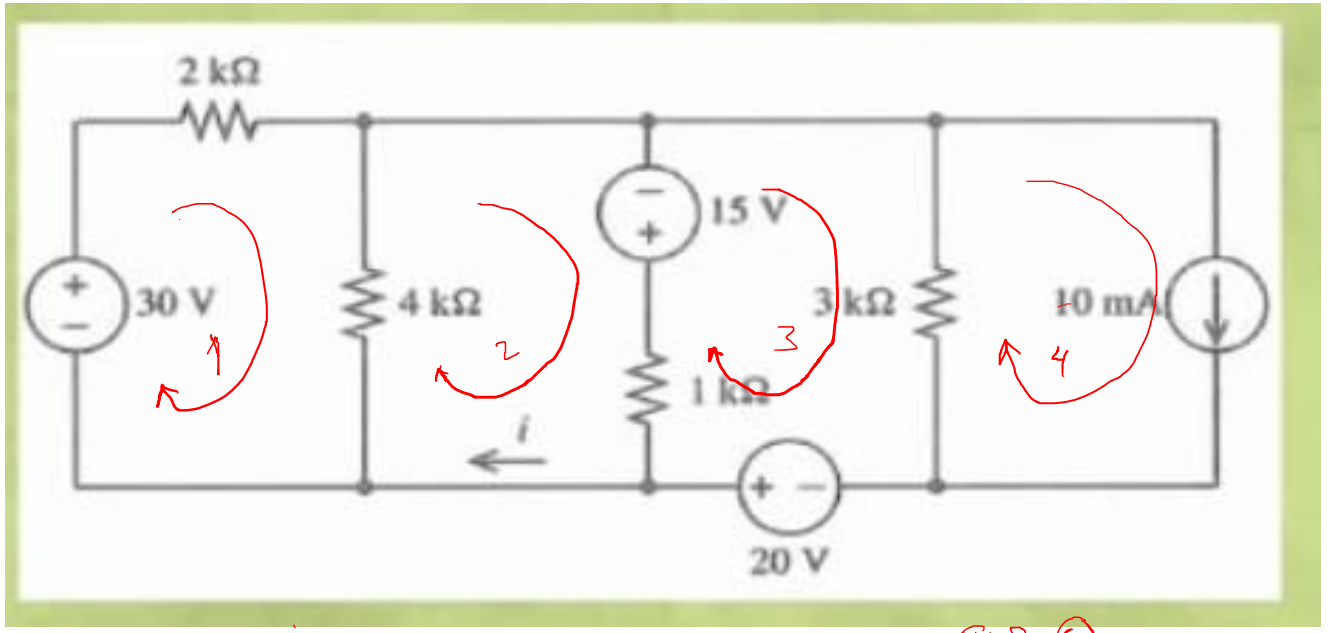
$i_{A4} = \frac{40}{100} \times 10 = 4\,A$

$i_4 = -i_{A4} = -4\,A$

$i_B = \frac{24}{64}(16\,A) = 6\,A$

$i_{B1} = \frac{200}{250} \times 6 = 4{,}8\,A$

$i_1 = -i_{B1} = -4{,}8\,A.$



Loop 1 =

$$-30 + 4(i_1 - i_2) + 2i_1 = 0$$

$$6i_1 - 4i_2 = 30$$

$$3i_1 - 2i_2 = 15 \quad \cdots (1)$$

Loop 2 =

$$4(i_2 - i_1) - 15 + 1(i_2 - i_3) = 0$$

$$-4i_1 + 5i_2 - i_3 = 15 \quad \cdots (2)$$

Loop 3 =

$$15 + 3(i_3 - i_4) - 20 + 1(i_3 - i_2) = 0$$

$$-i_2 + 4i_3 - 3i_4 = 5 \quad \cdots (3)$$

Loop 4 :

$$i_4 = 10 \text{ mA} \quad \cdots (4)$$

(4) → (3) → $-i_2 + 4i_3 - 30 = 5$

$$-i_2 + 4i_3 = 35 \quad \cdots (5)$$

$$4i_3 = 35 + i_2$$

$$i_3 = 35/4 + \frac{1}{4} i_2 \quad \cdots (5)$$

(5) → (2)

$$-4i_1 + 5i_2 - \frac{35}{4} - \frac{1}{4} i_2 = 15$$

$$-16i_1 + 20i_2 - 35 - i_2 = 60$$

$$-16i_1 + 19i_2 = 95 \quad \cdots (6)$$

(1) & (6)

$$3i_1 - 2i_2 = 15$$

$$-16i_1 + 19i_2 = 95$$

$$i_2 = \frac{\begin{vmatrix} 3 & 15 \\ -16 & 95 \end{vmatrix}}{\begin{vmatrix} 3 & -2 \\ -16 & 19 \end{vmatrix}} = \frac{285 + 240}{57 - 32}$$

$$= \frac{525}{25} = 21 \text{ mA}$$

$$i = i_2 = 21 \text{ mA}.$$